FLEET TEAM DIGEST

737MAX-FTD-32-18005

Issue Title : 737 MAX Nose Landing Gear (NLG) Steering Difficulties Due to Summing Mechanism Bearing Migration

Airplane Model	ATA	Minor Model(s)
737MAX	3251-52	-8, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
10/09/2018	10/14/2021	10/07/2021
Estimated Completion Date	**Next Update Date**	

Revision Description

Revision to SB 737-32-1544 released. All milestones complete, FTD closed.

Applicability

All 737 MAX aircraft prior to line number (L/N) 7027, except L/N 7023. Also applicable to L/N 7090, 7100, 7108 and 7113.

Description

This Fleet Team Digest (FTD) article informs operators of upper bearing migration events common to the early production Nose Landing Gear (NLG) summing mechanism assemblies, P/N 275A5351-1, utilized on 737 MAX aircraft (as shown in Figure 1 of the attachment). Additionally, this article summarizes the Root Cause/Corrective Action (RCCA) process and the resulting service bulletin.

Background

Three 737 MAX operators reported stiff/sluggish nose wheel steering followed by a rapid steering response during aircraft taxi operations. Flight crews also noted that the aircraft steering was not self-centering and rudder pedal steering input was ineffective. In all cases, the swaged P/N BACB10FU12J upper bearing (CMM 32-51-34, IPL Figure 1, Item 410) was found migrated out of the P/N 275A5358-1 bracket assembly (CMM 32-51-34, IPL Figure 1, Item 405) common to the P/N 275A5351-1 summing mechanism assembly (CMM 32-51-34, IPL Figure 1, Item 200), resulting in binding within the mechanism as shown in Figures 2 and 3 of the attachment. Four additional reports of summing mechanism binding were noted during the 737 MAX final assembly process.

Boeing initiated an RCCA investigation to fully understand and mitigate the problem. One of the failed in-service units was returned for a teardown evaluation. During this evaluation, primer/paint thicknesses were measured after which finishes were removed and the bearing, housing bore, and swage depth were measured. The bearing dimensions and housing bore were within

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drawing requirements. However, the swage depth was found to be outside the drawing specifications of 0.003-0.005 inches (the swage was not deep enough). As a result, the swage did not have sufficient retention capability to withstand the full spectrum of loads applied by the steering cables.

Following the RCCA investigation, Boeing and the component manufacturer agreed to make the following production changes to mitigate similar events on future 737 MAX aircraft:

1. The roller swaging of upper bearing will occur prior to paint/primer application on the bracket assembly. This will allow for an unimpeded swage integrity and depth measurement check. NOTE: Previously, the roller swage was applied and depth was measured after painting the bracket, which may have prevented an adequate swage check.
2. The design depth tolerance of the roller swage has been increased from 0.003-0.005 inches to 0.005-0.007 inches. This increases the bearing retention capabilities of the swage material.
3. A proof load test is performed on each summing mechanism assembly to ensure that the swage is capable of withstanding the full spectrum of possible loads (plus a factor of safety) without exhibiting bearing migration.

These changes have been incorporated on P/N 275A5351-1 summing mechanisms installed beginning with 737 MAX Line Number 7143. These units can be distinguished from potentially-suspect P/N 275A5351-1 summing mechanisms by the inclusion of one of the following part marks on the upper surface of the bracket assembly (as shown in Figure 4 of the attachment):

- Q.N. 264998

- Q.N. 265159

- Q.N. 8467589

- Date Code of 07/25/18 (25 July 2018) or later

To provide further configuration control, Boeing has released a new P/N 275A5351-2 summing mechanism, which incorporates all the aforementioned production changes. P/N 275A5351-2 is now being installed on all 737 MAX production aircraft beginning with L/N 7510.

Boeing issued service bulletin 737-32-1544 in March 2020. This service bulletin formalizes the on-wing inspections and replacement of the summing mechanism, depending on inspection findings. It also presents two options for using the updated P/N 275A5351-1 with specific markings or the new P/N 275A5351-2 summing mechanism which incorporates bearing retention improvements.

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Status

Boeing has released a revision (R01) to service bulletin 737-32-1544 to provide rework instructions to convert 275A5351-1 units to 275A5351-2 to avoid unnecessary scraping of summing mechanisms. In addition, the revised service bulletin provides instructions to re-part mark any 275A5351-1 units with a compliant date code as noted above to 275A5351-2. Boeing believes this will assist operators with better part configuration control.

Interim Action

Boeing recommends incorporatoration of service bulletin 737-32-1544 at a convenient maintenance opportunity to prevent possible operational disruptions.

Final Action

Production: Summing mechanism P/N 275A5351-2 is now being installed on all 737 MAX production aircraft beginning with L/N 7510.

Retrofit: Incorporation of SB 737-32-1544

Milestones

Root Cause Established: 31-JUL-2018 (Complete)

Solution Selected: 30-AUG-2018 (Complete)

Change Committed: 06-DEC-2018 (Complete)

Production Incorporation: 16-APR-2019 (Complete)

Parts Available: 05-SEP-2019 (Complete)

Service Bulletin Revision (R01): 30-SEP-2021 (Completed)

Operator Action

Boeing recommends that operators incorporate service bulletin 737-32-1544 which describes replacement of the original P/N 275A5351-1 (that fails the proof load test) with the new P/N 275A5351-2.

Part Information

NLG Summing Mechanism Theory:

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The NLG summing mechanism mixes steering control wheel input and nose gear position feedback to control the nose wheel steering metering valve. Steering wheel or rudder pedal input moves the summing mechanism from the neutral position. This makes an input to the steering metering valve module that sends pressure to the nose wheel steering actuators. The nose wheel steering actuators turn the nose landing gear wheels.

The movement of the nose gear wheels moves the summing mechanism back towards the neutral position. When the nose landing gear wheel position agrees with the steering wheel or rudder pedal position, the summing mechanism moves the metering valve to the neutral position. This stops the flow of hydraulic fluid to the nose wheel steering actuators.

The 737 MAX utilizes the newly designed summing mechanism, which incorporates a broken cable compensator in place of the summing bar (the 737NG steering system uses a one piece summing bar and does not include a broken cable compensator). The broken cable compensator prevents a steering command in the event of a loss in cable tension due to a cable break, loss of a pulley or bracket, or similar event.

The summing mechanism is above the upper steering plate on the front of the nose landing gear, as shown in Figure 1 of the attachment. On the aircraft, a cover blocks the summing mechanism from view. Lastly, this summing mechanism issue is unique to the 737 MAX installation and as such, only impacts 737 MAX aircraft.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	737-32-1544	New	02/28/2020
Aircraft Maintenance Manual (AMM)	32-51-52/401		
Illustrated Parts Catalog (IPC)	32-51-52-01, Item 175		
Component Maintenance Manual (CMM)	32-51-34		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
NLG Summing Mechanism Assembly (current)	275A5351-1		
Bracket Assembly (current)	275A5358-1		
NLG Summing Mechanism Assembly (improved)	275A5351-2		
Bracket Assembly (improved)	275A5358-3		
Upper Bearing	BACB10FU12J		

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Related Categories

- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Manufacturing Quality Issues : Issues related to airplane or component manufacturing quality

Attachments:

a) Summing Mechanism FTD Figures (REV A)

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